01 / PLATFORM ARCHITECTURE

Enterprise software for equipment, work and business operations.

MetraSlate connects equipment records to the people, customer work, technical evidence, documents and financial activity around them. Asset is the industry-independent business platform; Standard develops that foundation for an industry.

A browser workspace backed by transactional services

React and TypeScript provide the application interface, built with Vite. Kotlin services use Spring Boot on the JVM. PostgreSQL holds business and technical records; Flyway migrations version the schema. The browser requests operations, while the server determines scope, permission and the resulting transaction.

The records connect the complete job

A customer request can progress through an estimate, work order, equipment receipt, service, review, released document, delivery and invoice. These are linked records, not one status field repeated across departments. Individual instruments and job lines can retain their own progress and evidence.

One company boundary, with narrower access inside it

The application carries company context into database work. Laboratories and operating sites organize work within that company. A customer portal narrows access again to a customer account and, where configured, its permitted sites. A selected lab in the interface does not grant access to another company.

Structured data and document bytes have distinct homes

Operational values, relationships and document metadata live in PostgreSQL. Document storage addresses generated content by company and SHA-256 digest. Records retain the template and capture references needed to identify the document that was produced. A digest identifies content; it is not encryption or a substitute for access control.

What an enterprise buyer can evaluate

Follow one representative job across departments, examine the permission boundaries, inspect the technical record and issued document, then agree the deployment, migration and operating responsibilities for the organization.

02 / PRODUCTS AND DATA OWNERSHIP

One foundation. Different operating responsibilities.

Product selection determines the work the organization needs to run. It must also establish who owns the equipment data and which service relationships may see it.

Asset: equipment management and the business around it

Asset is industry-independent software for organizations that work with instrumentation and equipment. Its product scope combines asset management with general business operations such as customer work, purchasing, inventory, people and financial records. It does not prescribe the specialist calibration-provider workflow. Standalone Asset packaging is being developed from the shared platform.

Standard: that foundation developed for an industry

Standard adds specialized workflows, records and processes to the same asset and business foundation. The principal current example is the Metrology Edition: laboratory intake, structured calibration work, reference standards, review, certificates and quality records connected to the commercial job. The distinction is specialization, rather than an artificial removal of general business features from Asset.

Asset Light: focused equipment management

Asset Light provides asset management without the broader business operations. Current implementation work establishes an owner workspace, private fleet records, import, equipment history and supporting documents. Exact packaged features and rollout acceptance are confirmed during scoping; the existence of a shared platform capability alone does not include it in every package.

The customer portal remains a laboratory service relationship

The portal and the customer account inside a Standard laboratory refer to that customer's equipment in the lab's company. Portal access applies account and site permissions. It is not a renamed standalone Asset subscription, nor a separate copy of every record the lab shares with that customer.

Standalone owner data is not automatically shared

Asset and Asset Light use a private owner fleet. Explicit service exchange is planned to share the equipment selected for service and identify the receiving Standard company. One physical item may have separate records in several independent providers. Automatic fleet synchronization and that cross-company exchange are not represented here as completed functionality.

03 / IDENTITY, ACCESS AND ISOLATION

Enforce access where the records are read and changed.

Company isolation, staff authorization, customer visibility and commercial feature availability solve different problems. They are not interchangeable switches.

Authenticated identity and server-side policy

The application integrates WorkOS authentication with server-managed sessions. Request policies apply membership, role or privilege requirements at protected API operations. State-changing requests use a session anti-forgery token. Operator administration has a separate authorization path from employee and customer access.

Company context follows the database transaction

The tenant-aware connection layer applies company scope transaction-locally. Tenant-owned tables use PostgreSQL row-level security, with forced policies checking the company identifier for reads and writes. Missing required scope is rejected or fails closed. Platform-wide tables are a distinct class: their readers must apply explicit company predicates.

Customer and site permissions narrow the portal

Portal requests bind a customer account and its site grants. Unrestricted access and an empty permitted-site set have different meanings. A restricted contact cannot infer access to an unplaced instrument. For covered documents containing several equipment lines, the scope check withholds the complete document if a line falls outside that contact's grants.

Commercial availability differs from operational configuration

Commercial entitlements have company scope and validity windows. Module grants and tenant operational switches determine which available service groups are enabled. Neither replaces user permissions. Console-managed agreements can define customer terms; a complete independently priced feature catalog and per-feature enforcement remain separate product work.

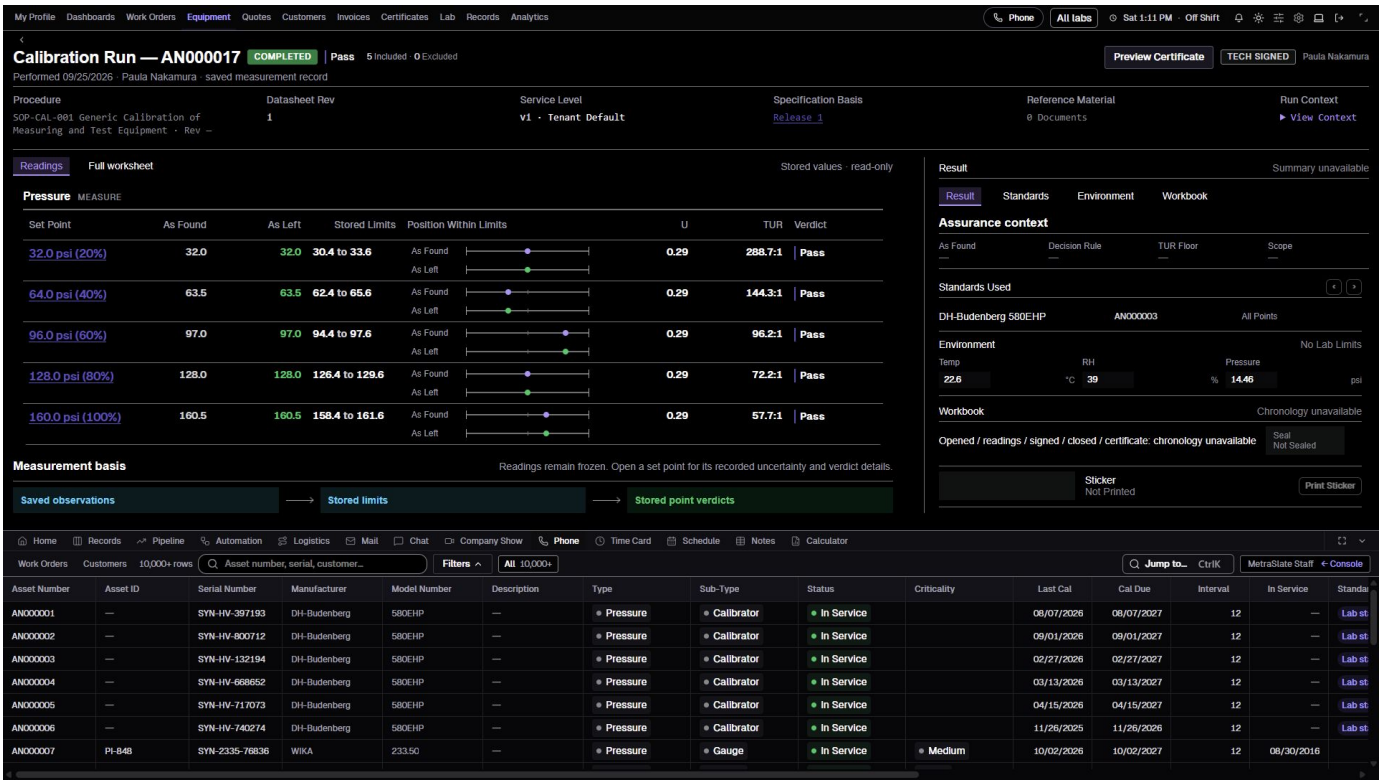
Browser and real-time channels need their own controls

The server supplies framing, content-type, referrer and opener headers, and HSTS on secure requests. Its Content Security Policy is report-only. Chat checks company and channel access for subscriptions; direct publishing to server broadcast destinations is rejected. Logical company isolation does not imply a dedicated database or encryption key per customer.

04 / STANDARD METROLOGY

Preserve the measurement and the basis of its result.

Standard connects the instrument and work order to structured readings, the evaluation basis, review and the issued document. The screenshot below is an authentic application capture with sample data.



Full calibration-record screen, shown without cropping. Select the image to open the original full-size capture.

Retain the recorded technical basis

Completed records preserve results, uncertainty, decision rules, standards and exclusions. Historical inspection reads that saved basis. Decimal values in the technical pipeline remain distinct from certificate display formatting. The frozen record and its correction lineage are described on the following page.

05 / HISTORY, DOCUMENTS AND ANALYTICS

Know what changed. Keep the evidence behind it.

Movement history, record activity and issued technical evidence answer different questions. MetraSlate keeps these meanings distinct instead of calling every timeline an immutable ledger.

Custody records physical handovers

Custody transfers identify the instrument, direction, counterparty and actor. An observed handover has a timestamp; a declared historical date remains a date rather than an invented instant. Transfers can distinguish a customer leg from outside-vendor service. A shared transactional writer connects the movement to the current recorded location and related workflow.

Inventory uses quantity movements

Stock receipts, consumption, transfers and adjustments are recorded as movements with their source and destination where relevant. Purchase-receipt and work references explain why the quantity moved. Corrections are additional adjustments rather than silent editing of the previous movement. This is a stock ledger, separate from the individual equipment service history.

Corrections preserve the earlier technical record

Frozen calibration data is protected by database controls. A correction links a successor and its reason to the earlier event. Technician signature, review and quality seal are distinct recorded actions. Record activity provides additional context, but does not imply identical append-only coverage for every field in every module.

Document versions connect design to issued output

Document Creator combines fixed content, typed record bindings and repeating tables. Validation checks bindings before publication. Generated documents retain template and capture references; frozen calibration and issuance records preserve the basis of released technical output. Previewing a template is a different action from issuing a customer document.

Automation and Analytics work from operational records

Workflow definitions describe events, conditions and actions, with execution records for investigation. Analytics exposes report definitions and contributing data through the reporting pipeline. Useful measures identify the period, scope and underlying records. A zero result, an unavailable measure and a failed read must remain different outcomes.

06 / HOSTING AND SERVICE CONNECTIONS

Define the deployment. Account for the dependencies.

The codebase includes an AWS deployment model and provider integrations. A configured design and an operating service are different evidence, so customer commitments refer to the environment being delivered.

AWS application and data services

Infrastructure definitions describe application containers behind an HTTPS load balancer, private-network PostgreSQL and S3 document storage. Database configuration includes KMS-encrypted storage and required TLS; document storage defines KMS encryption, versioning and public-access blocking. Actual applied settings are checked for the contracted environment.

Secrets, logs, monitoring and recovery

Deployment definitions cover managed secrets, application logging, alarms and database backup retention. Availability, retention and monitoring are environment-specific. Recovery must include both database records and the document bytes those records reference. Backup configuration is not evidence that a recovery exercise has succeeded.

External services are explicit operating dependencies

Authentication, mail, SMS, calling, maps and payment services each require their applicable provider configuration, permissions and commercial account. Their delivery, availability, cost and retention affect the implementation. A screen or installed SDK does not establish that a provider connection is enabled or accepted in a customer environment.

Billing and integration failures need visible outcomes

Billing processing records external subscription events and reconciles the local plan state. Recent implementation work separates plan-owned entitlements from independent grants and refuses unsupported subscription composition instead of guessing. Customer-specific pricing and feature availability are administered deliberately; an operational feature switch is not authorization to charge a customer.

Agree the actual service arrangement

Establish region, access ownership, subprocessors, retention, recovery objectives and incident responsibilities. Confirm enabled providers rather than assuming that map configuration or optional calling implies a live service. Availability and capacity commitments require evidence for the agreed deployment.

07 / IMPLEMENTATION AND TECHNICAL REVIEW

Prove the workflow your organization will rely on.

The technical buying process should connect enterprise requirements to the actual product configuration, migration and operating evidence. A demonstration is the beginning of that evaluation.

1. Establish product scope and record ownership

Choose asset-only, general business or industry-specialized scope. Define companies, sites, responsibilities and portal access. Identify private owner data and what a service provider may receive. Agree included capabilities and commercial terms.

2. Migrate representative records with reconciliation

Map identifiers, customers, sites, equipment, service dates, documents and relationships. Preserve the source and mapping decisions; reconcile counts and meaning with the customer. Imported certificates and declared dates remain distinct from calibrations performed inside Standard.

3. Exercise the complete job and its exceptions

Follow the selected package's work from request to outcome. Check partial progress, denied access, missing information, failed delivery and corrections. For Asset Light, validate owner equipment management rather than laboratory execution. The lab retains responsibility for its procedures and technical validation.

4. Review controls and the actual operating environment

Review isolation, role decisions, record integrity, provider configuration, recovery, monitoring and incident ownership. Define retention and export requirements. SOC 2 requires independent assurance evidence; an architecture overview is not a SOC 2 report or certification.

Start a technical walkthrough

Bring a representative asset list, work record, issued document and access model. We can then discuss the workflow, implementation and evidence relevant to your organization.

Request a demo: metraslate.com/demo

Technical and implementation discussion: metraslate.com/contact